

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.1 Bernoulli's Equation

Consider the x component of the momentum equation:

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \rho f_x + (F_x)_{viscous}$$

For an inviscid flow with no body forces:

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x}$$
$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} + \rho w \frac{\partial u}{\partial z} = -\frac{\partial p}{\partial x}$$

For steady flow:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$$

Multiply the above equation by dx:

$$u \frac{\partial u}{\partial x} dx + v \frac{\partial u}{\partial y} dx + w \frac{\partial u}{\partial z} dx = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx$$

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### 3.1 Bernoulli's Equation

Consider the streamline in three-dimensional space:

$$w dy - v dz = 0 \quad u dz - w dx = 0 \quad v dx - u dy = 0$$

We can further change this equation:

$$u \frac{\partial u}{\partial x} dx + u \frac{\partial u}{\partial y} dy + u \frac{\partial u}{\partial z} dz = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx$$

$$u \left( \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \right) = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx$$

The differential of  $u$  is:

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$

The equation can be written as :

$$u du = \frac{1}{2} d(u^2) = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx$$

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.1 Bernoulli's Equation

In a similar manner:

$$\frac{1}{2} d(v^2) = -\frac{1}{\rho} \frac{\partial p}{\partial y} dy$$

$$\frac{1}{2} d(w^2) = -\frac{1}{\rho} \frac{\partial p}{\partial z} dz$$

Thus, we have:

$$\frac{1}{2} d(u^2 + v^2 + w^2) = -\frac{1}{\rho} \left( \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \right)$$

$$u^2 + v^2 + w^2 = V^2$$

Due to the fact:

$$\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz = dp$$

The equation can be written as :

$$\frac{1}{2} d(V^2) = -\frac{dp}{\rho} \quad dp = -\rho V dV$$

### Euler's equation

It applies to an inviscid flow with no body forces, and it relates the change in velocity along a streamline  $dV$  to the change in pressure  $dp$  along the same streamline.

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.1 Bernoulli's Equation

**For incompressible flow:**

The above equation can be easily integrated between any two points 1 and 2 along a streamline.

$$\int_{p_1}^{p_2} dp = -\rho \int_{V_1}^{V_2} V dV$$

$$p_2 - p_1 = -\rho \left( \frac{V_2^2}{2} - \frac{V_1^2}{2} \right)$$

**Bernoulli's equation:**

$$p_1 + \frac{1}{2} \rho V_1^2 = p_2 + \frac{1}{2} \rho V_2^2$$

**Along a streamline:**

$$p + \frac{1}{2} \rho V^2 = \text{const}$$

**For an irrotational flow, the above equation is valid throughout the flow.**

# Chapter 3    Fundamentals of Inviscid, Incompressible Flow

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## 3.1 Bernoulli's Equation

### The physical significance:

When the velocity increases, the pressure decreases, and when the velocity decreases, the pressure increases.

The strategy for solving most problems in inviscid, incompressible flow is as follows:

1. Obtain the velocity field from the governing equations.
2. Once the velocity field is known, obtain the corresponding pressure field from Bernoulli's equation.

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.2 Pressure Coefficient

Pressure, by itself, is a dimensional quantity. The pressure coefficient is defined as:

$$C_p \equiv \frac{p - p_\infty}{q_\infty} \quad \text{where} \quad q_\infty = \frac{1}{2} \rho_\infty V_\infty^2$$

**For incompressible flows**,  $C_p$  can be expressed in terms of velocity only. Consider the flow over an aerodynamics body immersed in a freestream with pressure  $p_\infty$  and velocity  $V_\infty$ . Pick an arbitrary point in the flow where the pressure and velocity are  $p$  and  $V$ , respectively. From Bernoulli's equation

$$p_\infty + \frac{1}{2} \rho V_\infty^2 = p + \frac{1}{2} \rho V^2 \quad \text{or} \quad p - p_\infty = \frac{1}{2} \rho (V_\infty^2 - V^2)$$

**The pressure coefficient:**

$$C_p \equiv \frac{p - p_\infty}{q_\infty} = \frac{\frac{1}{2} \rho (V_\infty^2 - V^2)}{\frac{1}{2} \rho V_\infty^2} = 1 - \left( \frac{V}{V_\infty} \right)^2$$

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.3 Laplace's Equation

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \vec{V} = 0$$

For incompressible flow,  $\rho = \text{constant}$

$$\frac{\partial \rho}{\partial t} = 0 \quad \nabla \cdot (\rho \vec{V}) = \rho \nabla \cdot \vec{V}$$
$$\nabla \cdot \vec{V} = 0$$

For irrotational flow:

$$\vec{V} = \nabla \phi$$
$$\nabla \cdot (\nabla \phi) = 0$$
$$\nabla^2 \phi = 0$$

**Laplace's equation.** Solutions of Laplace's equation are called harmonic functions.



# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.3 Laplace's Equation

Laplace's equation is written below in terms of the three common orthogonal coordinate systems:

**Cartesian coordinates:**

$$\phi = \phi(x, y, z)$$

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

**Cylindrical coordinates:**

$$\phi = \phi(r, \theta, z)$$

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

**Spherical coordinates:**

$$\phi = \phi(r, \theta, \Phi)$$

$$\nabla^2 \phi = \frac{1}{r^2 \sin \theta} \left[ \frac{\partial}{\partial r} \left( r^2 \sin \theta \frac{\partial \phi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{\partial}{\partial \Phi} \left( \frac{1}{\sin \theta} \frac{\partial \phi}{\partial \Phi} \right) \right] = 0$$



# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.3 Laplace's Equation

1. Any irrotational, incompressible flow has a velocity potential and stream function that both satisfy Laplace's equation.
2. Conversely, any solution of Laplace's equation represents the velocity potential or stream function ( two-dimensional) for an irrotational, incompressible flow.

Note that Laplace's equation is a second-order linear partial differential equation. The fact that it is **linear** is particularly important, because the sum of any particular solutions of a linear differential equation is also a solution of the equation:

$$\phi = \phi_1 + \phi_2 + \cdots + \phi_n$$

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.3 Laplace's Equation

The general approach to the solution of irrotational, incompressible flows can be summarized as follows:

1. Solve Laplace's equation for  $\Phi$  or  $\psi$  along with the proper boundary conditions.
2. Obtain the flow pressure from  $\vec{V} = \nabla \phi$  or  $u = \partial \psi / \partial y$   $v = -\partial \psi / \partial x$
3. Obtain the pressure from Bernoulli's equation,

$$p + \frac{1}{2} \rho V^2 = p_{\infty} + \frac{1}{2} \rho V_{\infty}^2$$

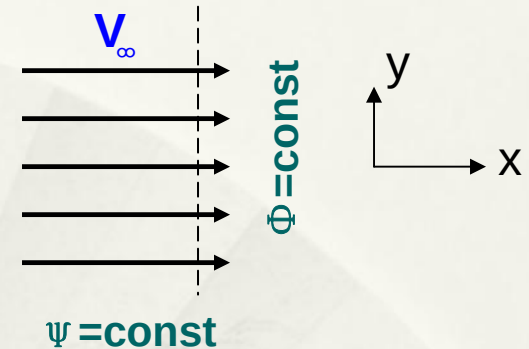
Where  $p_{\infty}$  and  $V_{\infty}$  are known freestream conditions.

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

## 3.4 Uniform Flow

Consider a uniform flow with velocity  $V_\infty$  oriented in the positive  $x$  direction.

$$\frac{\partial \phi}{\partial x} = u = V_\infty \quad \frac{\partial \phi}{\partial y} = v = 0$$



Integrate the above equation with respect to  $x$ , we have

$$\phi = V_\infty x + f(y) \quad \text{where } f(y) \text{ is a function of } y \text{ only}$$

Integrate the above equation with respect to  $y$ , we have

$$\phi = \text{const} + g(x) \quad \text{where } g(x) \text{ is a function of } x \text{ only}$$

Since  $\Phi$  is the same function.

$$\phi = V_\infty x + \text{const}$$

### 3.4 Uniform Flow

Note that in a practical aerodynamic problem, the actual value of  $\Phi$  is not significant; rather,  $\Phi$  is always used to obtain the velocity by differentiation; that is

$$\vec{V} = \nabla \phi$$

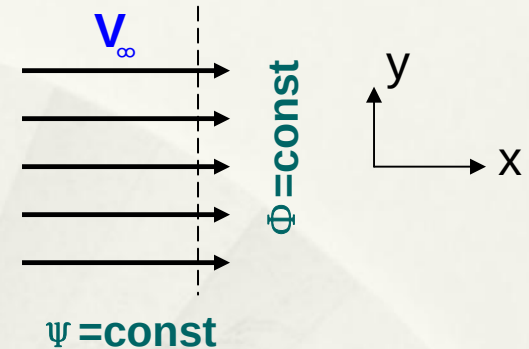
The velocity potential for a uniform flow with velocity  $V_\infty$  oriented in the positive x direction.

$$\phi = V_\infty x$$

Now, consider the incompressible stream function  $\psi$

$$\frac{\partial \psi}{\partial y} = u = V_\infty \quad \frac{\partial \phi}{\partial x} = -v = 0$$

The stream function for a uniform flow with velocity  $V_\infty$  oriented in the positive x direction.



$$\psi = V_\infty y$$

### 3.4 Uniform Flow

Circulation

$$\Gamma = - \oint_C \vec{V} \cdot d\vec{s}$$

$$\Gamma = - \vec{V}_\infty \cdot \oint_C d\vec{s}$$

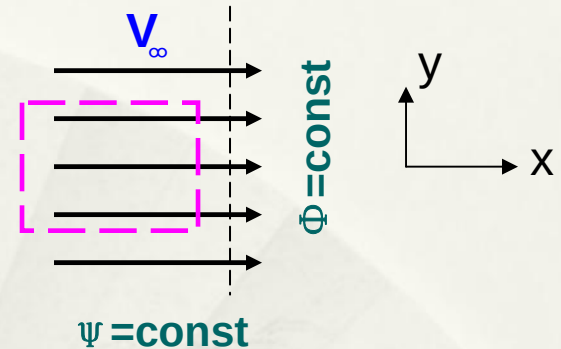
$$\Gamma = \vec{V}_\infty \cdot 0 = 0$$

or

$$\Gamma = - \iint_S (\nabla \times \vec{V}) \cdot d\vec{S}$$

$$\nabla \times \vec{V} = 0$$

$$\Gamma = 0$$



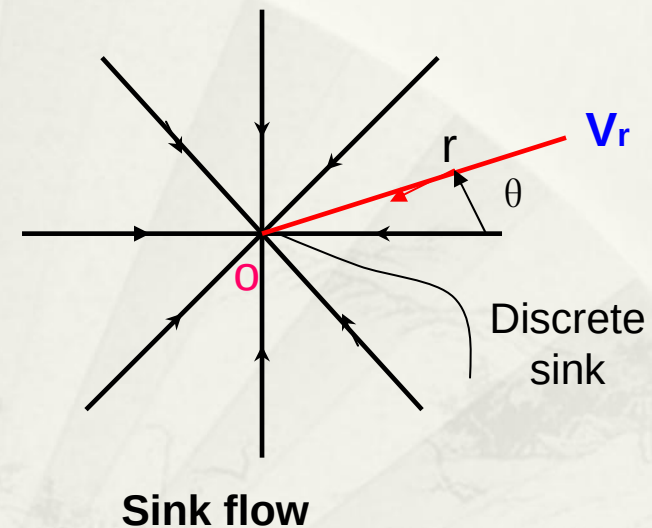
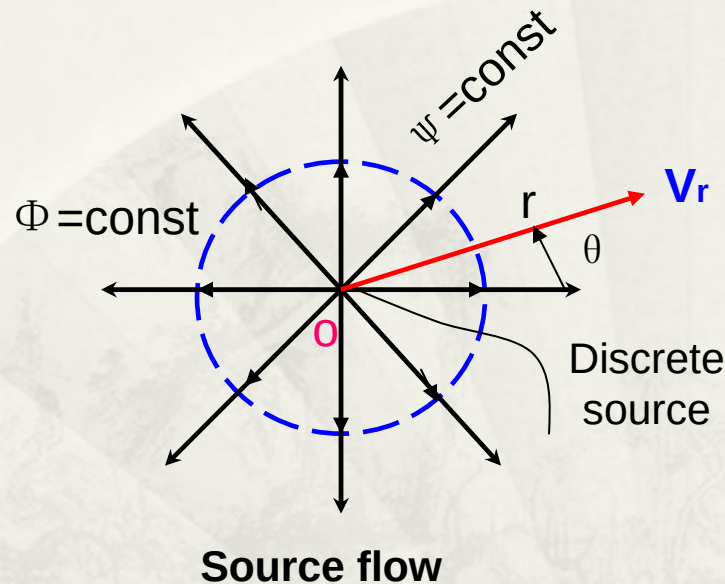
The result is consistent

(The uniform flow is irrotational)

**Conclusion:** Circulation around any closed curve in a uniform flow is zero

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

## 3.5 Source and Sink Flows



### Source flow:

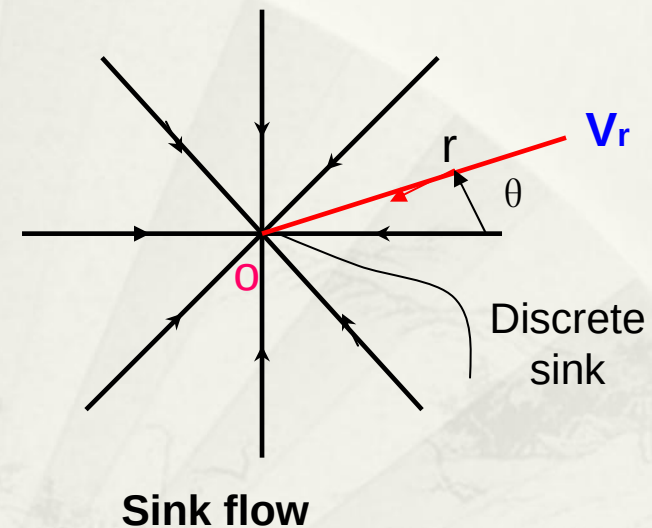
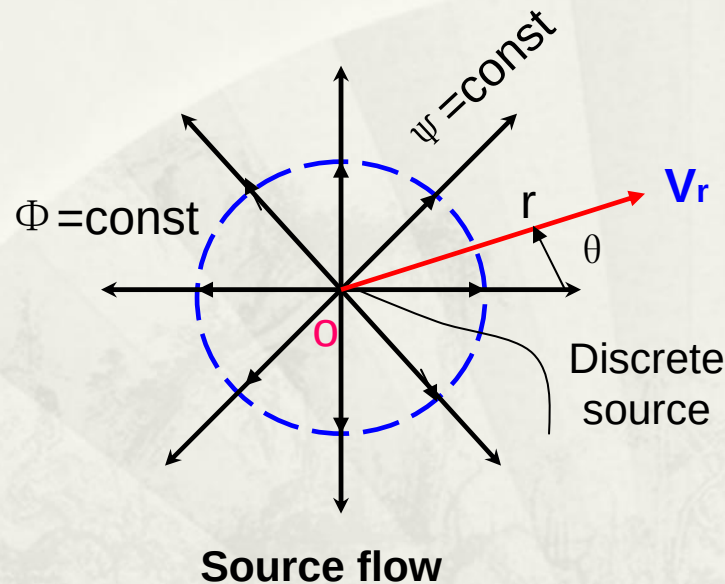
Consider a two-dimensional, incompressible flow where all the streamlines are straight lines emanating from a central point  $O$ .

Moreover, let the velocity along each of the streamlines vary inversely with distance from point  $O$ .



# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

## 3.5 Source and Sink Flows



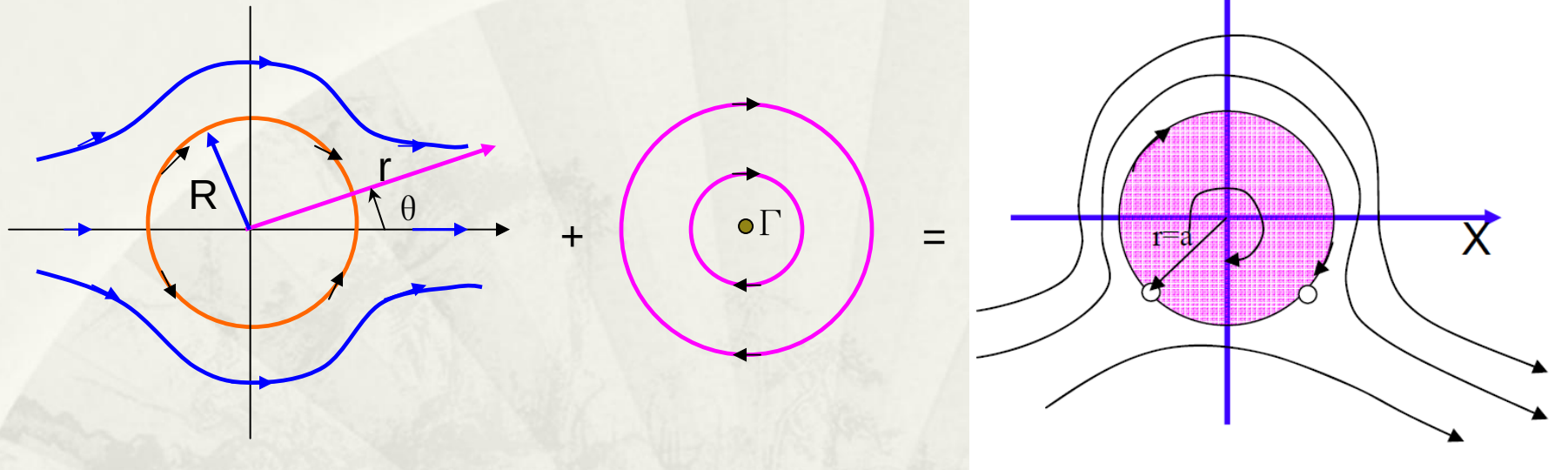
### Source flow:

1. Source flow is a physically possible incompressible flow, that is  $\nabla \cdot \vec{V} = 0$ , at every point except the origin, where  $\nabla \cdot \vec{V}$  becomes infinite.
2. Source flow is irrotational at every point.



## Chapter 3 Fundamentals of Inviscid, Incompressible Flow

### 3.8 Lifting flow over a cylinder



Flow satisfies continuity at every point  $r \geq R$ .

$$\nabla \cdot \vec{V} = 0$$

Flow satisfies irrotationality at every point  $r \geq R$ .

$$\nabla \times \vec{V} = 0$$

## Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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### 3.8 Lifting flow over a cylinder

#### Force:

Force on an elemental distance on the surface of the cylinder:

$$d\vec{F} = -p_b R d\theta \hat{e}_r$$

$$d\vec{F} = -p_b R d\theta (\cos \theta \hat{i} + \sin \theta \hat{j})$$

$$\vec{F} = \int_0^{2\pi} -p_b R d\theta (\cos \theta \hat{i} + \sin \theta \hat{j})$$

The drag per unit span is:

$$D' = \vec{F} \cdot \hat{j} = \int_0^{2\pi} -p_b \cos \theta R d\theta$$

The lift per unit span is:

$$L' = \vec{F} \cdot \hat{i} = \int_0^{2\pi} -p_b \sin \theta R d\theta$$

## Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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### 3.8 Lifting flow over a cylinder

#### Force:

As we know, in incompressible flow the total pressure  $p_0 = p + \frac{\rho V^2}{2}$  which is a constant throughout the flow.

There is no flow normal to the surface,  $V_r=0$ .

$$p_b = p_0 - \frac{\rho(V_r^2 + V_\theta^2)}{2} = p_0 - \frac{\rho V_\theta^2}{2} \quad (V_r = 0)$$

$$p_b = p_0 - \frac{\rho}{2} \left( -2V_\infty \sin \theta - \frac{\Gamma}{2\pi R} \right)^2$$

$$p_b = p_0 - 2\rho V_\infty^2 (\sin \theta)^2 - \rho V_\infty \sin \theta \frac{\Gamma}{\pi R} - \frac{\rho \Gamma^2}{8\pi^2 R^2}$$

## Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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### 3.8 Lifting flow over a cylinder

#### Force:

The drag per unit span is:

$$D' = R \int_0^{2\pi} -\left(p_0 - 2\rho V_\infty^2 (\sin \theta)^2 - \rho V_\infty \sin \theta \frac{\Gamma}{\pi R} - \frac{\rho \Gamma^2}{8\pi^2 R^2}\right) \cos \theta d\theta = 0$$

d' Alembert's paradox still prevails.

The lift per unit span is:

$$L' = R \int_0^{2\pi} -\left(p_0 - 2\rho V_\infty^2 (\sin \theta)^2 - \rho V_\infty \sin \theta \frac{\Gamma}{\pi R} - \frac{\rho \Gamma^2}{8\pi^2 R^2}\right) \sin \theta d\theta = \rho V_\infty \Gamma$$

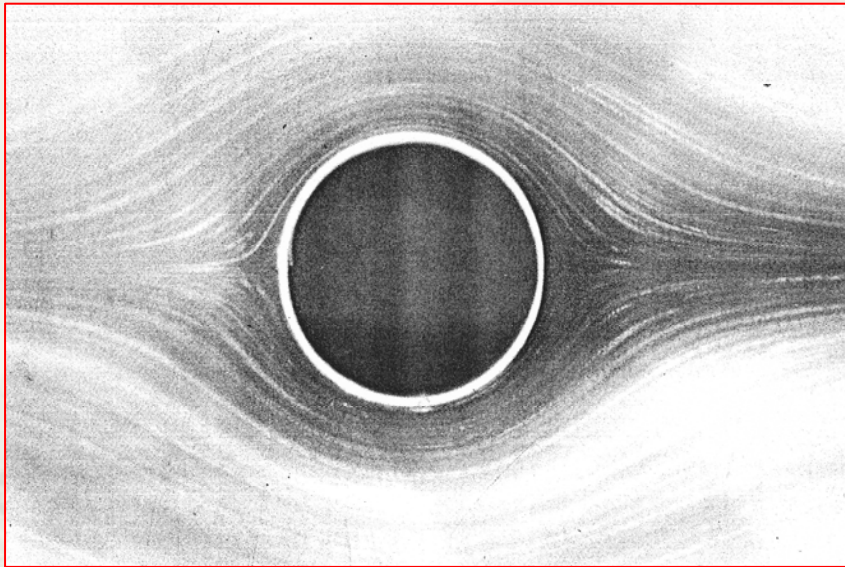
Kutta-Joukowski theorem

In inviscid, incompressible flow, the resultant force per unit span acting on a 2-D body of any cross section is equal to  $\rho V_\infty \Gamma$  and acts perpendicular to  $V_\infty$ .

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

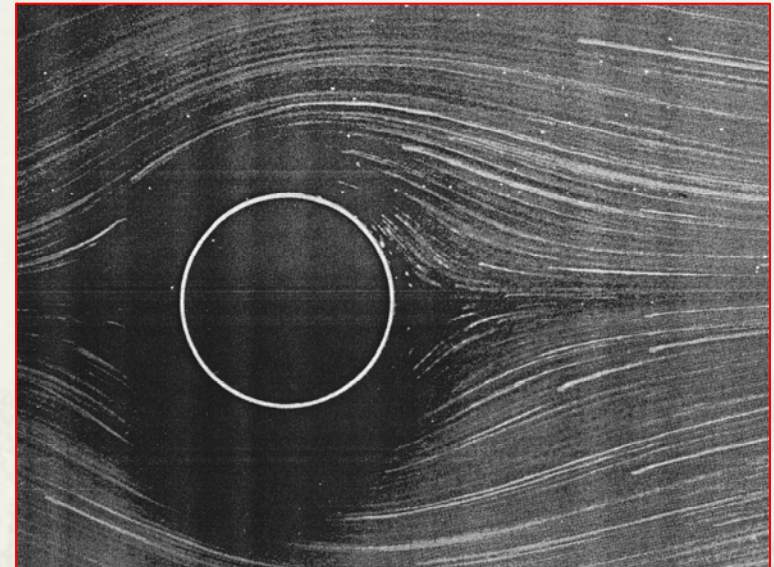
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## 3.9 Visualization



Circular cylinder at  $Re=0.16$

The flow past a solid body is reversible and hence symmetric about the cylinder.



Circular cylinder at  $Re=1.54$

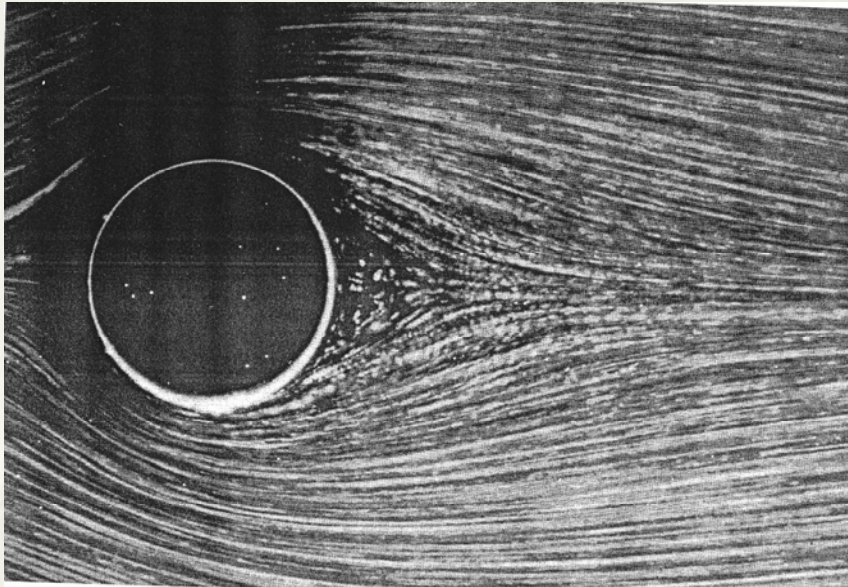
The flow has clearly lost the fore-and-aft symmetry of the figure on the left.



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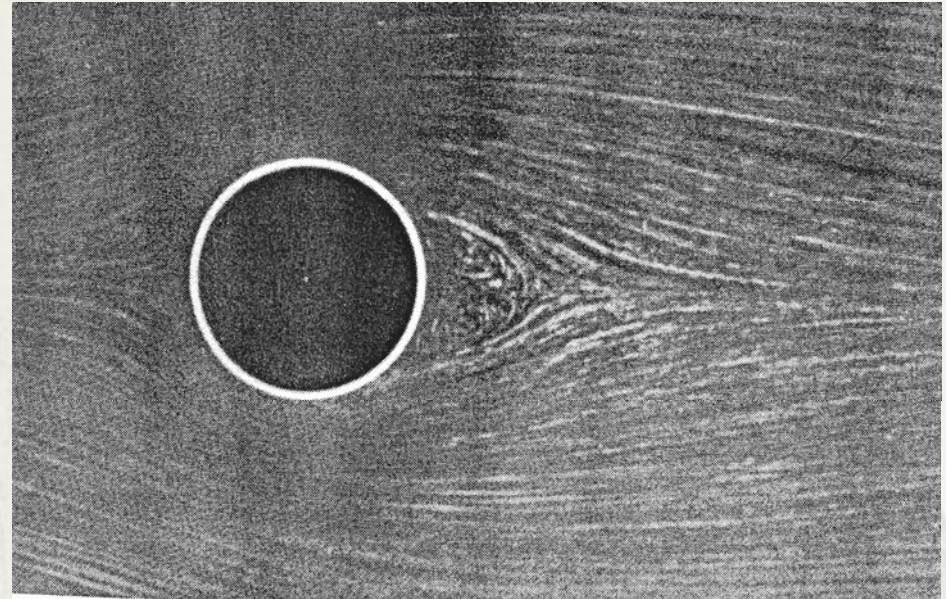
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## 3.9 Visualization



Circular cylinder at  $Re=9.6$

The flow has clearly separated to form a pair of re-circulating eddies.



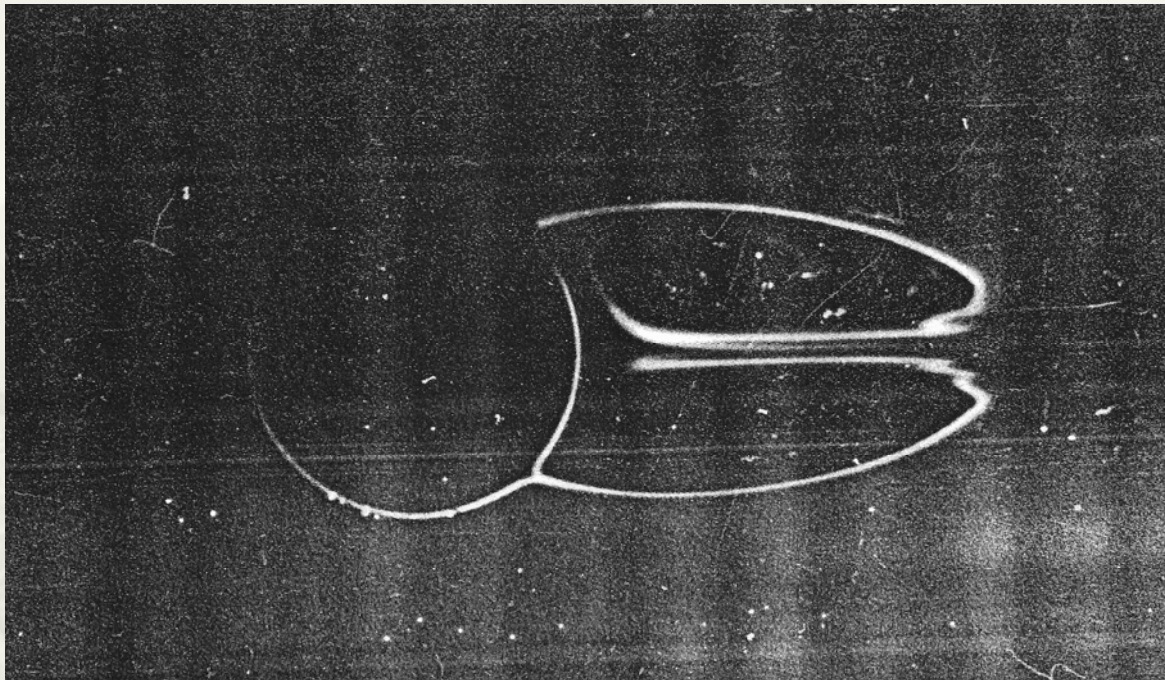
Circular cylinder at  $Re=13.1$

The standing eddies become elongated in the flow direction as the speed increases.

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.9 Visualization



Circular cylinder at  $Re=28.4$

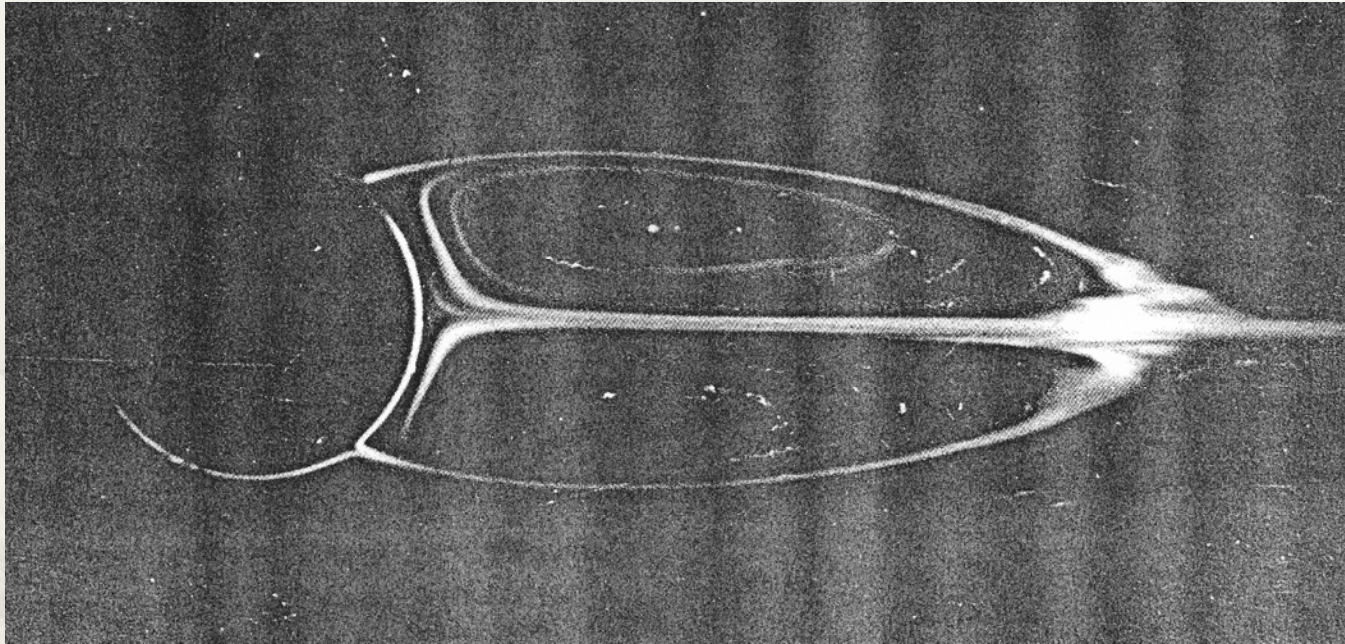
Here just the boundary of the recirculating region has been made visible by coating the cylinder with condensed milk and setting it in motion through water.



# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.9 Visualization

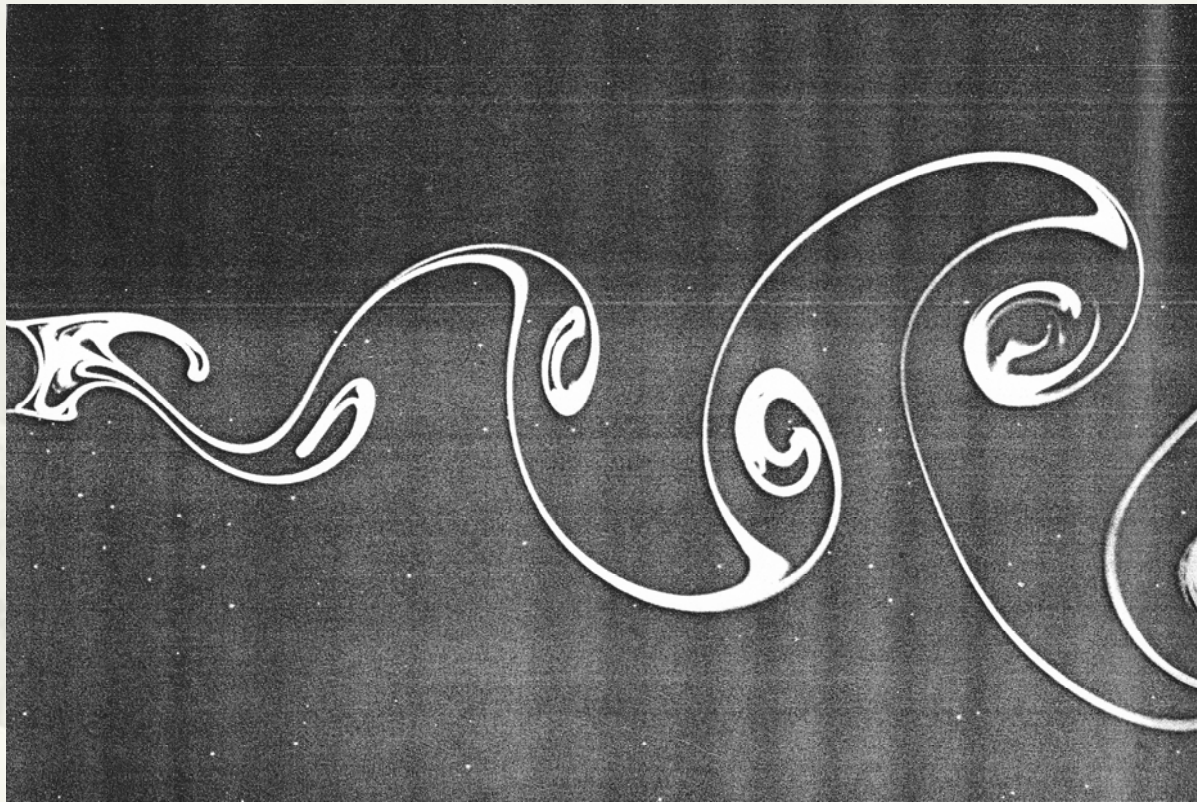


Circular cylinder at  $Re=41.0$

This is approximate upper limit for steady flow. Far downstream the wake has already begun to oscillate sinusoidally.

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### 3.9 Visualization

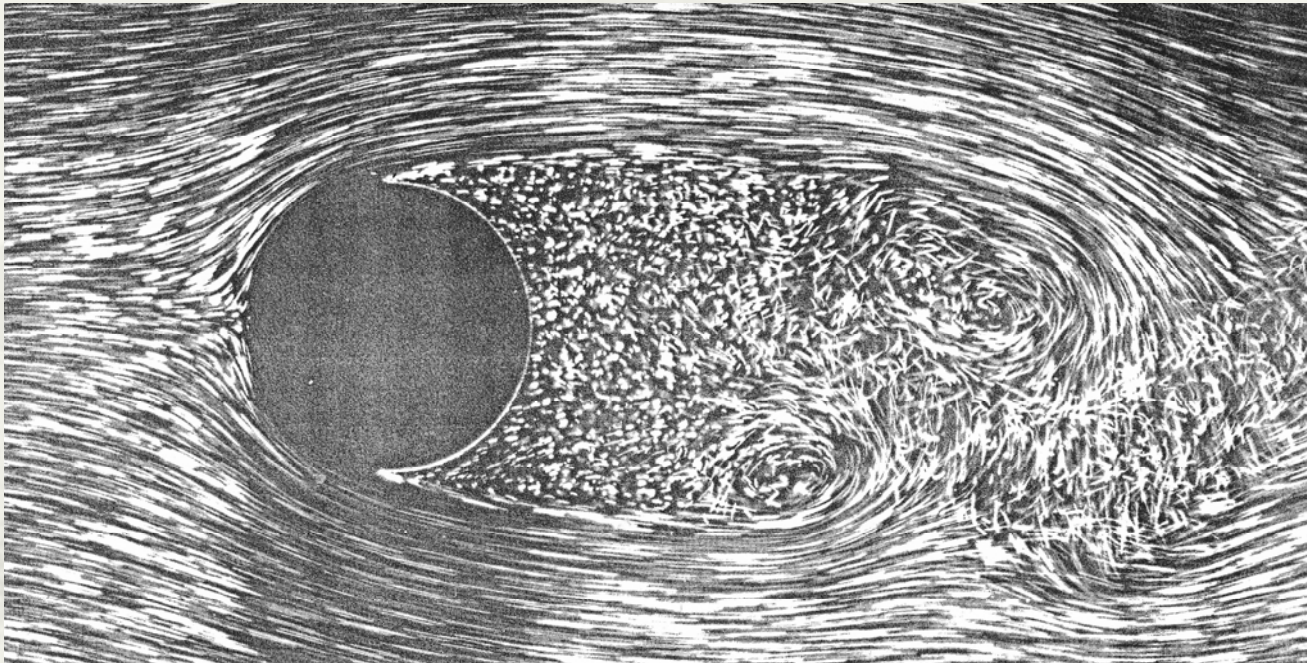


Kármán vortex street behind a circular cylinder at  $Re=140$



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## 3.9 Visualization

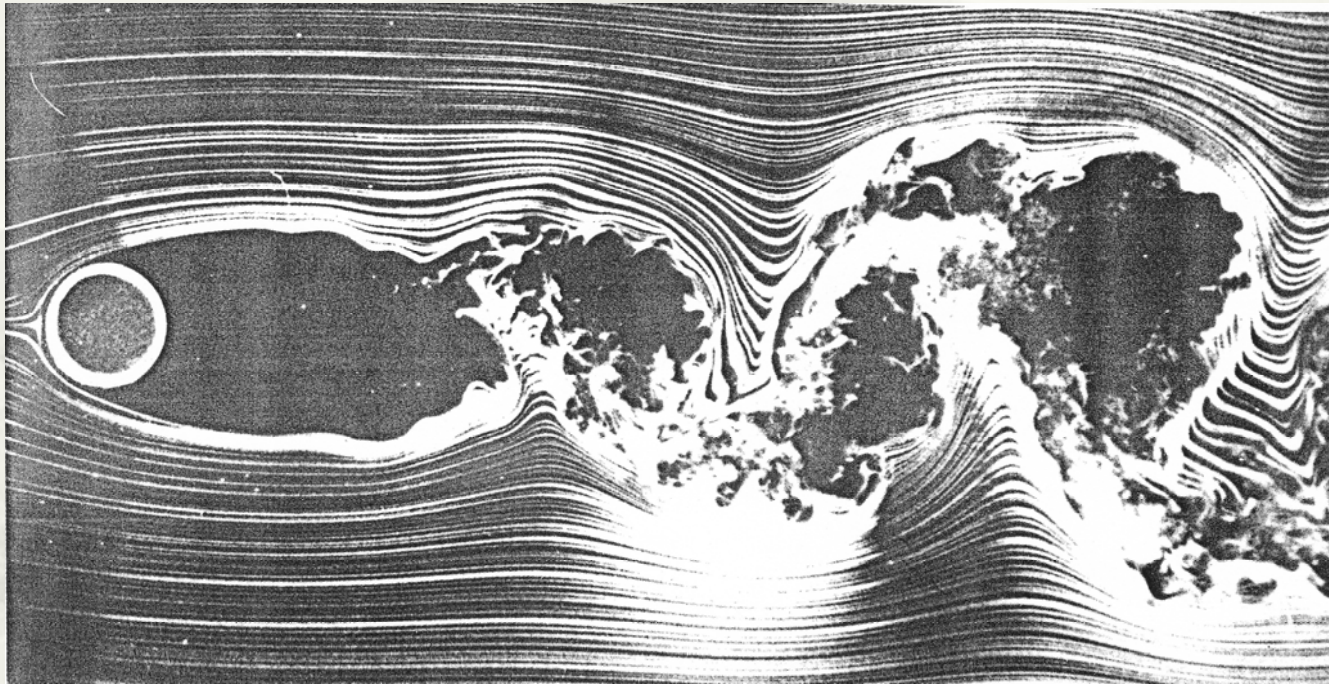


Circular cylinder at  $Re=2000$

The flow is laminar over the front, separates, and breaks up into a turbulent wake.

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## 3.9 Visualization



Circular cylinder at  $Re=10,000$

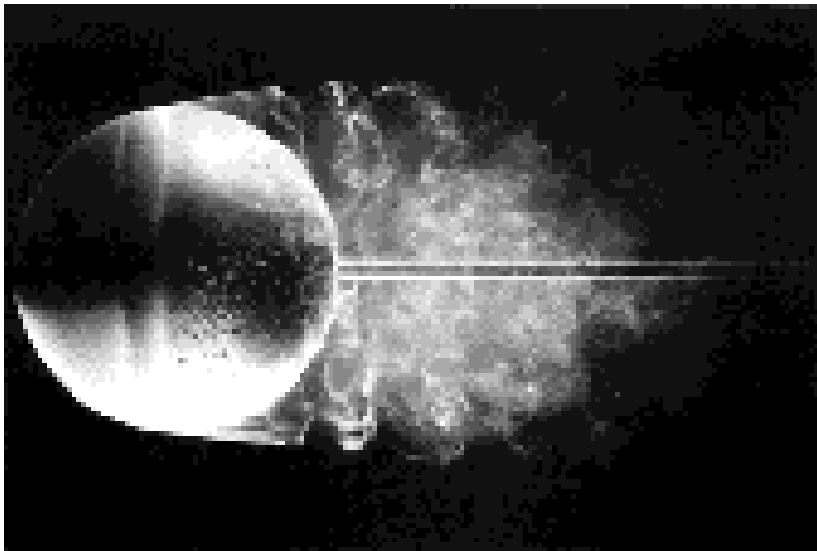
At five times the speed of the photograph in the last slide, the flow pattern is scarcely changed.

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.9 Visualization

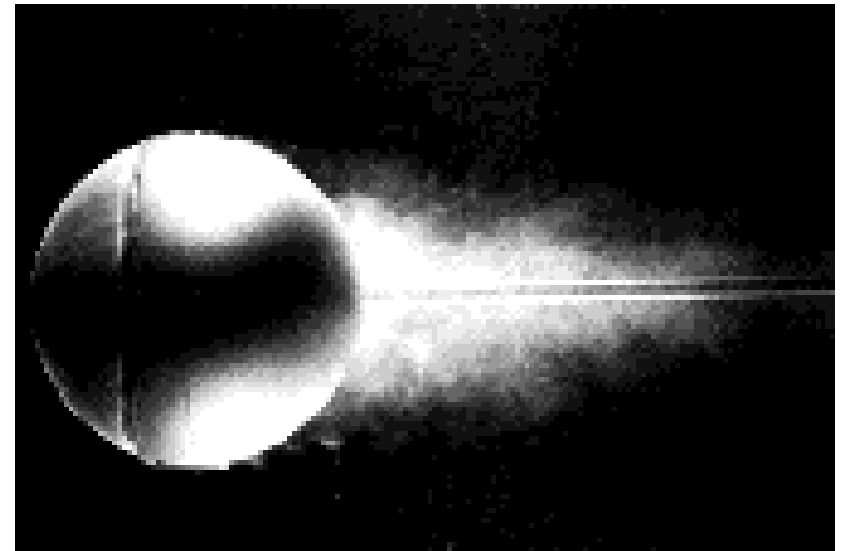
### Flow over a sphere



*a*

**Reynolds number = 15,000**

Dye in water shows a laminar boundary layer separating ahead of the equator and remaining laminar for almost one radius.



*b*

**Reynolds number=30,000**

A wire loop ahead of the equator trips the boundary layer. It becomes turbulent, so that it separates farther rearward than if it were laminar.



# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.10 Summary

### Bernoulli's equation

$$p + \frac{1}{2}\rho V^2 = \text{const}$$

- (a) Applies to inviscid, incompressible flows only.
- (b) Holds along a streamline for a rotational flow.
- (c) Holds at every point throughout an irrotational flow.
- (d) In the form given above, body forces (such as gravity) are neglected, and steady flow is assumed.

### Pressure coefficient

$$C_p = \frac{p - p_\infty}{q_\infty}$$

Dynamic pressure:  $q_\infty = \rho V_\infty^2 / 2$

For incompressible steady flow with no friction:

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2$$

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.10 Summary

### Elementary flows

| Type of Flow                | Velocity                                                                                                  | $\Phi$                                 | $\psi$                                  |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------|
| Uniform flow in x direction | $u = V_{\infty}$                                                                                          | $V_{\infty} x$                         | $V_{\infty} y$                          |
| Source                      | $V_r = \frac{\Lambda}{2\pi}$                                                                              | $\frac{\Lambda}{2\pi} \ln r$           | $\frac{\Lambda}{2\pi} \theta$           |
| Vortex                      | $V_{\theta} = -\frac{\Gamma}{2\pi}$                                                                       | $-\frac{\Gamma}{2\pi} \theta$          | $\frac{\Gamma}{2\pi} \ln r$             |
| Doublet                     | $V_r = -\frac{k}{2\pi} \frac{\cos \theta}{r^2}$<br>$V_{\theta} = -\frac{k}{2\pi} \frac{\sin \theta}{r^2}$ | $\frac{k}{2\pi} \frac{\cos \theta}{r}$ | $-\frac{k}{2\pi} \frac{\sin \theta}{r}$ |

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.10 Summary

### Inviscid flow over a cylinder

#### (a) Nonlifting (uniform flow and doublet)

$$\psi = (V_{\infty} r \sin \theta) \left(1 - \frac{R^2}{r^2}\right)$$

where  $R = \text{radius of cylinder} = k/2\pi V_{\infty}$

Surface velocity:

$$V_{\theta} = -2V_{\infty} \sin \theta$$

Surface pressure coefficient:

$$C_p = 1 - 4 \sin^2 \theta$$

$$L = D = 0$$

# Chapter 3 Fundamentals of Inviscid, Incompressible Flow

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## 3.10 Summary

### Inviscid flow over a cylinder

(b) Lifting (uniform flow + doublet + vortex)

$$\psi = (V_{\infty} r \sin \theta) \left(1 - \frac{R^2}{r^2}\right) + \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

where  $R = \text{radius of cylinder} = k/2 \pi V_{\infty}$

Surface velocity:

$$V_{\theta} = -2V_{\infty} \sin \theta - \frac{\Gamma}{2\pi R}$$

Lift per unit span:

$$L' = \rho_{\infty} V_{\infty} \Gamma$$

$$D = 0$$



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## 3.10 Summary

### Governing equations

$$\nabla \bullet \vec{V} = 0$$

(Condition of incompressibility)

$$\nabla^2 \phi = 0$$

(Laplace's equation; holds for irrotational, incompressible flow)

or  $\nabla^2 \psi = 0$

### Kutta – Joukowski theorem

For a closed two-dimensional body of arbitrary shape, the lift per unit span is :

$$L' = \rho_{\infty} V_{\infty} \Gamma$$